

Introduction to Ab Initio Nuclear Theory

Lecture 3: Nuclear forces and how to fake them

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More man-hours have been given to this problem than to any other scientific question in the history of mankind.
– H. Bethe 1953

1 Introduction

In this Lecture we will discuss the nuclear force, that is the interaction between nucleons. To this day, the exact structure of the nuclear force remains *unknown*. In principle it originates from the residual interaction between the quarks that make up the nucleons. In practice, it remains impossible to determine it as such and that has created the incentive for nuclear physicists to develop more or less approximate descriptions of it (and dedicate entire Lectures to it). Developing models for the nuclear interaction remains an active area of research in nuclear physics and various approaches have been developed to attack the problem. Many descriptions have emerged as consequential over the past century, they have been adopted widely by many-body theorists often creating sub-communities in their own right. More importantly, often small variations between different descriptions of the nuclear force lead to largely different descriptions of many-nucleon systems: a clear manifestation of the strongly interacting nature of nucleons in the scientific literature.

A proper treatment of all the influential descriptions of the nuclear force would likely require many more lectures in addition to a proper introduction to the field-theoretic tools often employed in the interaction-making business. Instead we will cast a divide on the literature, define Phenomenological and Ab initio forces (though the nomenclature is just the names that these forces often go by and it's still a subject of very interesting debate), and give an overview of some widely used examples from each side. Before we start we will provide an overview of the effective range expansion which is an essential piece in the construction of all microscopic models of the nuclear force. Finally, we will discuss a nuclear manifestation of serendipity: how the strong interaction between nucleons that makes many-nucleon systems notoriously hard to describe, gives a way out by allowing the engineering of nucleon-nucleon interactions between cold atoms via the so-called Feshbach resonances.

The following exposition is not historical; it's not even reverse-historical. The various forces are ordered to provide intuition. The naming of various terms is kept as consistent with the literature as possible. Finally note that we will focus on the nuclear force part of the nucleon-nucleon potentials, while their treatment of the electromagnetic part might often differ, we will not address this.

2 The effective range expansion

The purpose of this section is not a formal introduction to two-body scattering but rather to provide some intuition into how a two-body interactions affects the wavefunction of the two-body system. More importantly, the aim will be to see how these contributions change as the potential is probed at different energies, or momenta.

The time-independent Schrödinger's equation for the relative part of a two-body system at the centre of mass is

$$-\frac{\hbar^2}{2m_r}\nabla^2\psi(\mathbf{r}) + V(\mathbf{r})\psi(\mathbf{r}) = E\psi(\mathbf{r}) \quad (1)$$

where the total wavefunction would be $\Psi(\mathbf{r}_1, \mathbf{r}_2) = e^{i\mathbf{P}\cdot\mathbf{R}/\hbar}\psi(\mathbf{r})$ with $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$ and $\mathbf{R} = (\mathbf{r}_1 + \mathbf{r}_2)/2$, and the reduced mass defined as

$$m_r = \frac{m_1 m_2}{m_1 + m_2} \quad (2)$$

In the center-of-mass frame we can view the problem as a one-particle of mass m_r scattering off the potential $V(r)$. Assuming that the particle is incident from the z -axis, the solution for ψ should be axially symmetric and so it's

useful to expand as

$$\psi(\mathbf{r}) = \sum_{l=0}^{\infty} A_l P_l(\cos \theta) R_l(r) \quad (3)$$

where A_l are coefficients to be determined under the right boundary conditions, P_l is the l -th Legendre polynomial, and R_l satisfies the radial equation

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR_l}{dr} \right) + \left[k^2 - \frac{l(l+1)}{r^2} - \frac{2m_r}{\hbar^2} V(r) \right] R_l = 0 \quad (4)$$

Outside the potential's range, i.e., where $V(r) = 0$, R_l becomes a linear combination of Henkel functions and their complex conjugates which is

$$R_l(r) \propto \cos \delta_l(k) j_l(kr) - \sin \delta_l(k) n_l(kr) \quad (5)$$

where j_l and n_l are Bessel and Neumann functions, and δ_l is the so-called phase-shift of R_l which is determined so that R_l 's behaviour outside the potential's range matches its behaviour inside at the boundary of the two regions. In other words, δ_l carries the information of how the wavefunction outside the potential's range changes depending on what happened inside it and crucially, it is momentum (or energy) dependent. At large kr , R_l outside the range of the potential can be written as

$$R_l(r) \approx \frac{1}{kr} \sin \left(kr - \frac{\pi}{2} l + \delta_l(k) \right) = \frac{1}{2ikr} \left[e^{i(kr - \frac{\pi}{2} l + \delta_l)} - e^{-i(kr - \frac{\pi}{2} l + \delta_l)} \right] \quad (6)$$

At the same time, the wavefunction of a particle coming from the z -axis and being scattered by a potential at the origin can be written, at large distance r ,

$$\psi_k(\mathbf{r}) = e^{ikz} + f_k(\theta) \frac{e^{ikr}}{r} \quad (7)$$

where $f(\theta)$ is called the *scattering amplitude*. Plugging Eq. (6) into Eq. (3) and comparing it with Eq. (7) we can determine the coefficients A_l so that the wavefunction will have the right behaviour at the asymptotic limit:

$$A_l = (2l+1) i^l e^{i\delta_l(k)} \quad (8)$$

As expected, the information of what happened within the potential's range, carried by δ_l , persists at the asymptotic limit which in principle is much further than the potential's range. The same comparison also provides us with the scattering amplitude,

$$f_k(\theta) = \frac{1}{2ik} \sum_{l=0}^{\infty} (2l+1) \left(e^{2i\delta_l(k)} - 1 \right) P_l(\cos \theta) . \quad (9)$$

This tells us that δ_l uniquely defines the scattering amplitude, reiterating that δ_l carries the information of what happened inside the potential's range. It's important to distinguish that δ_l doesn't carry *all* the scattering information but only the one that makes it out of the potential's range.

A scattering experiment, like the ones used to constrain the nuclear force and that we'll mention later, can measure the differential cross-section,

$$\frac{d\sigma_k}{d\Omega} = |f_k(\theta)|^2 \quad (10)$$

or the total cross section

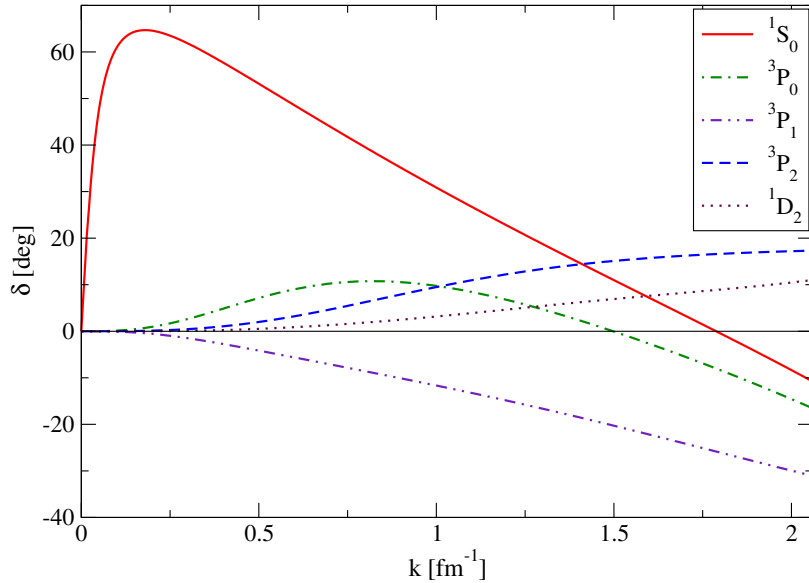
$$\sigma_k = \int d\Omega \frac{d\sigma}{d\Omega} = \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l . \quad (11)$$

Thus, these experiments can infer the phase-shifts which, again, carry the information of how the scattering that happened at $kr \sim 1$ affects the wavefunction at large distances $kr \gg 1$.

As has already been explicit, the phase shift depends on the momentum k , that is, the energy of the incident particles. This means that the wavefunction is affected differently at different energies. One then would like to have a systematic expansion that organizes these contributions in some order: this could answer the question "what more (or less) do I learn about the interaction that drives the scattering by increasing the energy of the incident beam?" As it turns out, the appropriate function to expand is not the phase-shift itself but the cotangent of it which yields the so-called effective-range expansion

$$k^{2l+1} \cot[\delta_l(k)] = -\frac{1}{a^{(l)}} + \frac{1}{2} r_e^{(l)} k^2 + \sum_{n=0}^{\infty} (-1)^{n+1} c_n^{(l)} k^{2n+4} . \quad (12)$$

This expansion has an additional implication which we'll explore further later. It says that processes that are driven by low-energy scattering (such processes are common in nuclear physics), only care about specific features of the two-particle interaction. In turn, any potential that possess these features can be used to model these processes and the neglected features introduce errors that can be estimated from the next term in the expansion. In other words, and as we'll see later, 12 represents an effective theory for the two-particle interaction. Below we plot the measured phase-shifts of np -scattering.



3 General properties of the nuclear force

Before we dive into different models for the nuclear interaction, let's start from what we expect to be true from the general properties of two-body interactions and experimental facts.

In general we expect the nuclear interaction to be an operator acting on space (coordinate or momentum), the particle spins, and the particle isospins. Knowing that the Pauli matrices augmented with the identity matrix are the basis of all operators acting on spin-1/2 and isospin-1/2 states, the general nuclear interaction can be expressed as

$$V = \sum_{\substack{ij=0 \\ kl=0}}^3 V_{ijkl}(\mathbf{x}_1, \mathbf{x}_2, \mathbf{v}_1, \mathbf{v}_2) \sigma_i^{(1)} \sigma_j^{(2)} \tau_k^{(1)} \tau_l^{(2)} \quad (13)$$

The properties of the nuclear interaction that we immediately expect are:

Hermeticity The potential should be a Hermitian operator.

Permutation invariance Invariance under the exchange of particle indices is assumed for any interaction of indistinguishable particles.

Translational invariance The potential should only depend on relative coordinates.

Galilean invariance The potential is unaffected by a transformation to a system of reference moving with finite speed.

Time-reversal invariance The potential shouldn't depend on the direction of the arrow of time.

Rotational invariance in coordinate space Invariance under rotation in space restricts the types of terms that the potential could include, both in terms of $\mathbf{x}$ and $\mathbf{p}$ but also in terms of the spin operators (or Pauli matrices).

Three terms are expected: a central term, a tensor term, and a spin-orbit term The expectation of a central term, that is, a term that depends only on the separation of the nucleons is standard. The expectation for a tensor term is expected from the quadrupole moment of the deuteron which can't be explained by pure central forces. Finally a spin-orbit term is expected as it's related to the spin-orbit term in the average single-particle nuclear potential that is essential for the reproduction of the magic numbers in nuclei.

Charge (in)dependence and charge symmetry By definition, *charge independence* refers to the potential invariance under rotations in isospin space (e.g., a neutron and a proton are connected by an isospin rotation). *Charge symmetry* is the special case where a potential is invariant under a rotation by 180 about the y -axis in isospin space (when the positive z -direction is associated with the positive charge). *Charge independence breaking (CIB)* or *charge dependence (CD)* is a breaking of the charge symmetry: any rotation in isospin space changes the potential. In the special case where a rotation by 180 about the y -axis in isospin space changes the potential, we call that charge dependence charge symmetry breaking (CSB). Obviously, CSB is a special case of CIB (or CD). Note that these concepts *do not* refer to the Coulomb interaction, but only to the nuclear part of the nucleon-nucleon interaction, i.e., after electromagnetic effects have been removed. Thus CIB means different interactions in nn , pp , and np scattering while, CSB means different interactions between nn and pp scattering. The presence of these effects can be seen and quantified in the scattering lengths of each channel:

$$a_{pp} = -17.3 \text{ fm} , \quad r_{e \text{ } pp} = 2.85 \text{ fm} \quad (14)$$

$$a_{np} = -23.74 \text{ fm} , \quad r_{e \text{ } np} = 2.77 \text{ fm} \quad (15)$$

$$a_{nn} = -18.9 \text{ fm} , \quad r_{e \text{ } pp} = 2.75 \text{ fm} \quad (16)$$

That is, we can define a length scale for CSB and CIB,

$$\Delta a_{\text{CSB}} = a_{pp} - a_{nn} = 1.6 \text{ fm} , \quad (17)$$

$$\Delta r_{\text{CSB}} = r_{e \text{ } pp} - r_{e \text{ } nn} = 0.1 \text{ fm} , \quad (18)$$

$$\Delta a_{\text{CIB}} = a_{pp} - a_{nn} = 6.44 \text{ fm} , \quad (19)$$

$$\Delta r_{\text{CIB}} = r_{e \text{ } pp} - r_{e \text{ } nn} = 0.08 \text{ fm} , \quad (20)$$

On a fundamental (high-resolution) level, and at the current understanding, CIB is caused by the mass difference of the up and down quarks inside the nucleons and electromagnetic interactions among the quarks. On the nucleon level, these lead to mass differences between hadrons of the same isospin multiplet, meson mixing, and irreducible meson-photon exchanges.

4 Phenomenological two-nucleon and three-nucleon forces

4.1 Two-nucleon forces

4.1.1 The meson-exchange approach: The Charge-Dependent Bonn nucleon-nucleon (CD Bonn) potential

A major group of models for the nuclear force is the so-called meson-exchange models. These models describe the nuclear force as originating from one- and many-boson exchanges between the nucleons where the bosons are mesons with masses below the nucleon mass. (Remember that mesons are bosons made of two quarks: a quark and an antiquark). These mesons (with their masses and couplings to the nucleon) are the pions ($\pi^{0,+,-}$, $m_\pi \approx 140 \text{ MeV}$, $g_\pi^2/4\pi = 13.6$), the eta mesons (η , $m_\eta = 550 \text{ MeV}$, $g_\eta^2/4\pi \leq 0.4$), rho mesons (ρ , $m_\rho \approx 770 \text{ MeV}$, $g_\rho^2/4\pi = 0.84$), and omega mesons (ω , $m_\omega \approx 782 \text{ MeV}$, $g_\omega^2/4\pi \approx 20$). A full description of the two-nucleon scattering using meson exchanges was achieved by the so-called full Bonn model. However, deriving multi-meson exchanges was very involved and the resulting potentials were energy dependent. These practical and conceptual problems prohibited the full Bonn potential from being widely used in nuclear structure calculations. Instead, a one-meson exchange version of it was developed, named the *Charge Dependent (CD) Bonn potential*.

The Charge-Dependent Bonn (CD Bonn) nucleon-nucleon potential was a one-boson-exchange potential that described the nuclear force as originating from light ($m \leq M$) meson exchanges. On top of the mesons mentioned above, π , η , ρ , and ω , two scalar-isoscalar (spin and isospin $S = 0$ and $T = 0$, respectively) σ bosons were included. The σ -mesons are hypothetical and were introduced to account for the lack of 2π and $\pi\rho$ exchanges, as we'll discuss later. Finally, due to its small coupling the η meson was dropped; it was later determined that in the full Bonn model, a good fit of the NN data actually favored a vanishing η coupling.

Note that as will be the case for almost all of our discussion, the nucleon was considered structureless, that is, a point particle. The leading contribution of nucleon's internal structure in the nucleon nucleon scattering is the nucleon's lowest excited state, the Δ resonance with mass $m_\Delta = 1232 \text{ MeV}$, usually named $\Delta(1232)$. Keeping with CD Bonn's philosophy of only keeping degrees of freedom with mass lower than the nucleon's, the $\Delta(1232)$ resonance and other heavier degrees of freedom arising from the nucleon's structure were not included. This automatically means that the resulting potential could not describe NN scattering up to an arbitrary energy where the neglected degrees of freedom could be activated. In other words, the potential had to be regulated. This is a concept that we will encounter a lot in this Lecture and it lies at the heart of developing good descriptions of the nuclear force. There is many ways to regulate interactions, that is, to ensure that your potential is not imposing physics in energy scales where it shouldn't work, and in the case of the CD Bonn this was done by including form factors to the meson-nucleon vertices. These are momentum dependent functions that "kill" the meson-nucleon interaction beyond some momentum scale in the momentum exchanged between the nucleon and the meson. Note that this is not a

peculiarity of nuclear interactions, but in any theory the choice of degrees of freedom restricts the validity of the interactions between them to the energy range in which their neglected structure is irrelevant.

Providing the final expressions for the entire potential is pointless as it will span a few pages. Instead we will a) examine the term arising from the pion, the so-called one-pion-exchange potential (OPE) and b) phenomenologically/qualitatively describe the contribution of each meson, i.e., what functional term does each additional meson append to the potential). Dissecting the OPE will allow us the luxury of squinting at the rest of the terms while providing a decent point of contact with other simplified models of the nuclear force and a gateway to the Chiral forces later where pion exchanges reappear. The qualitative description of each meson's contributions will build on the intuition gained by the OPE.

The contribution of each meson is calculated by deriving a scattering amplitude. For the pion this is

$$V_\pi(\mathbf{q}', \mathbf{q}) = -\frac{g_\pi^2}{4M^2} \frac{(E' + M)(E + M)}{(\mathbf{q} - \mathbf{q}')^2 + m_\pi^2} \left(\frac{\boldsymbol{\sigma}_1 \cdot \mathbf{q}'}{E' + M} - \frac{\boldsymbol{\sigma}_1 \cdot \mathbf{q}}{E + M} \right) \left(\frac{\boldsymbol{\sigma}_2 \cdot \mathbf{q}'}{E' + M} - \frac{\boldsymbol{\sigma}_2 \cdot \mathbf{q}}{E + M} \right), \quad (21)$$

where $E = \sqrt{\mathbf{q}^2 + M^2}$, $E' = \sqrt{\mathbf{q}'^2 + M^2}$ and $\mathbf{q}$ the relative nucleon momenta of the initial and final scattering states. This is the full OPE term, i.e., the part of the nuclear interaction originating from the exchange of a pion between nucleons. Note that we haven't enforced $E = E'$, as one would expect from the conservation of energy. This is related to the so-called off-shell behaviour of the potential which we'll discuss later. For now if we enforce $E = E'$ and also take the non-relativistic limit $E \approx M$, we arrive at

$$V_\pi(\mathbf{k}) = -\frac{g_\pi^2}{4M^2} \frac{(\boldsymbol{\sigma}_1 \cdot \mathbf{k})(\boldsymbol{\sigma}_2 \cdot \mathbf{k})}{\mathbf{k}^2 + m_\pi^2}, \quad (22)$$

where $\mathbf{k} = \mathbf{q}' - \mathbf{q}$ is the momentum exchange. This is called the static approximation and it might be the one you're most familiar with. Indeed, performing the Fourier transform to coordinate space, we get

$$V_\pi(\mathbf{r}) = \frac{g_\pi^2}{12\pi} \left(\frac{m_\pi}{2M} \right)^2 \left[\left(\frac{e^{-m_\pi r}}{r} - \frac{4\pi}{m_\pi^2} \delta^{(3)}(\mathbf{r}) \right) \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 + \left(1 + \frac{3}{m_\pi r} + \frac{3}{(m_\pi r)^2} \right) \frac{e^{-m_\pi r}}{r} \mathbf{S}_{12} \right], \quad (23)$$

where $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$, the relative distance between the nucleons. This is the OPE exchange used by the first approximations to the nuclear force as it naturally provides the tensor force. This approximation is also called the local approximation because as we'll discuss later, a dependence on $\mathbf{k}$ that was induced by making $E = E' = M$ means that the potential is local in coordinate space, i.e., depends only on $\mathbf{r}$.

Equipped with how the OPE looks like, onwards to the rest of the mesons: η , ρ , and ω and their utility. As we saw explicitly, the pion provides the tensor force; this is modified at short range by the ρ meson. The ω provides the a spin-orbit term and the two σ bosons are responsible for the intermediate-range attraction. The existence of the σ -mesons was controversial at best at the time and the need for their inclusion was a major drawback in previous models. However, it was observed that their effect was similar to that of a two-pion exchange contribution. Conversely, in the full Bonn Model, including π , 2π and ω exchanges describes the peripheral partial waves of the the NN scattering well but provides unnaturally large binding at shorter distances; remember peripheral partial waves are those with high angular momenta and hence, in a semi-classical picture, they probe the interaction at large distances. A $\pi\rho$ exchange appeared to solve the intermediate distance overbinding (note this is the effect of a two-boson exchange which is conceptually distinct from that of two one-boson exchanges; nevertheless, the effect is somewhat similar to the ρ 's reduction of the π 's tensor force at short distances mentioned above).

Charge (in)dependence and charge symmetry Within the meson-exchange approach, at the OBE level, the CSB due to the difference in the neutron and proton masses accounts for about 15% of the observed difference in the scattering lengths. This mass splitting enters through the kinetic energies and their effect on the OBE diagrams. On the TBE level, the CSB can be fully explained by these effects. There the largest contribution comes from

the 2π exchanges with those including $N\Delta$ intermediate states contributing the most. Because of this outstanding contribution to the CSB by TBE terms, the CD Bonn potential includes it.

Within the meson-exchange approach, the CIB is caused mainly by the difference in masses of the various pions. At the multi-mesons exchange level, the largest contribution, $\sim 50\%$, comes from the OPE with the 2π -exchange following. Pinning the CIB to the pion also means that because of its low mass (which translates to larger range which in turn translates to effects in high partial waves), CIB effects are expected in all partial waves.

On-shell and off-shell behaviours of the nuclear potential All our preceding discussion was centred on a static picture: starting with the two-body scattering and the effective range expansion, we have been considering nuclear forces acting once between scattering particles which we then intercept at infinity investigating transitions in their states before and after the scattering. This approach hides some of the properties of the nuclear interaction that come into play when the scattering particles are very close. In equivalent terms, by investigating transitions between free-particle states that conserve the total energy we can't see what is happening when particles are not described by one single state or their properties do not conserve the energy. In the literature the part of the nuclear potential that describes these scenarios is called *off-shell* inheriting the name from Quantum Field Theory where particles are said to scatter on(off)-shell when the total momentum before and after the collision is (not) conserved. In coordinate space the same effect results in non-local forces.

Nucleons are bound in nuclei and so their scattering is often off-shell. This means that the off-shell behaviour of nuclear potentials affects the nuclear structure determined from them, but it cannot be seen in the NN phase-shifts. Thus in practice, phase-shift equivalent potentials dictate different nuclear structure.

4.1.2 The AV_n potential (AV4, AV8, AV14, AV18,...)

The idea of the AV18 potential was a different approach to the meson-exchange theory: what if we don't restrict the form of the operators to the OBE? As mentioned before, turning the full Bonn model into a potential was hard and impractical, so CD-Bonn was restricted to (mostly) OBE terms and since no systematic inclusion of beyond-OBE terms was attempted, the resulting potential is restrictive in form and not systematically improvable. The AV18 potential attempted to include *all reasonably expected* local operator forms into a potential and fit it to experimental data. Here what we mean as reasonably expected is terms that don't break any symmetry of the interaction that we know to be true (see above). Many potentials were created in this logic and named based on the number of operators included, e.g., 4 operators in AV4, 8 operators in AV8, etc. From this family of potentials we will focus on AV18 as the most widely used one, however, note that the other family members are also still in use.

The form of the AV18 interaction is schematically,

$$V(\mathbf{r}_1, \mathbf{r}_2) = V_{\text{EM}}(\mathbf{r}_1, \mathbf{r}_2) + V_{\text{OPE}}(\mathbf{r}_1, \mathbf{r}_2) + V_{\text{R}}(\mathbf{r}_1, \mathbf{r}_2) \quad (24)$$

where the terms are, in order, the EM interaction, the one-pion-exchange term, and a remainder term which is the one that contains the 18 operators. Due to the importance of the pion-exchange in the phenomenological explanation of the nuclear force, that OBE term was kept as a leading contribution and combined with 18 more terms. Note that as we'll see later explicitly the terms in Eq. (24) should be seen as spin and isospin operators. Ignoring the EM term, and differing most of the discussion on the OPE to the previous section, we'll simply look at the OPE term as it contains relevant definitions and we'll focus on the remainder term. The OPE is taken as

$$V_{\text{OPE},pp} = f_{pp}^2 V_{\pi}(r; m_{\pi^0}) \quad (25)$$

$$V_{\text{OPE},np} = f_{pp} f_{nn} V_{\pi}(r; m_{\pi^0}) + 2(-1)^T f_c V_{\pi}(r; m_{\pi^{\pm}}) \quad (26)$$

$$V_{\text{OPE},nn} = f_{nn}^2 V_{\pi}(r; m_{\pi^0}) , \quad (27)$$

where

$$V_\pi(r; m) = \left(\frac{m}{m_{\pi^\pm}} \right)^2 \frac{mc^2}{3} [Y_\mu(r) \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j + T_\mu \mathbf{S}_{ij}] , \quad (28)$$

where

$$Y_\mu(r; c) = \frac{e^{-\mu r}}{\mu r} \left(1 - e^{-cr^2} \right) , \quad (29)$$

$$T_\mu(r; c) = \left[1 + \frac{3}{\mu r} + \frac{3}{(\mu r)^2} \right] \frac{e^{-\mu r}}{\mu r} \left(1 - e^{-cr^2} \right)^2 , \quad (30)$$

where $\mu = mc/\hbar$. These are the same functions used for the traditional OPE only multiplied with an exponential cutoff.

The remainder term is

$$V_R(\mathbf{r}_1, \mathbf{r}_2) = V_{ST}^c(r) \mathbf{1} + V_{ST}^{l2}(r) \mathbf{L}^2 + V_{ST}^t(r) \mathbf{S}_{12} + V_{ST}^{ls}(r) \mathbf{L} \cdot \mathbf{S} + V_{ST}^{ls2}(r) (\mathbf{L} \cdot \mathbf{S})^2 . \quad (31)$$

You can see now that $V_R(\mathbf{r}_1, \mathbf{r}_2)$ is an operator and it contains a central term (V^c , i.e., not an operator), a term depending on the particles' total orbital angular momentum (V^{l2}), a tensor term (V^t), a spin-orbit term (V^{ls}), and a squared spin-orbit term (V^{ls2}). The indices ST signify that these functions are allowed to be different for different spin-isospin channels, e.g., the terms for the deuteron are different than the ones for s-wave neutron-neutron scattering). For each term ($i = c, l2, t, ls, ls2$) the form factors are decomposed as

$$V_{ST}^i(r) = I_{ST}^i T_\mu^2(r) + [P_{ST}^i + (\mu r) Q_{ST}^i + (\mu r)^2 R_{ST}^i] W(r) \quad (32)$$

where $\mu = (c/\hbar) \sum_i m_{\pi^i}/3 = (c/\hbar)(m_{\pi^0} + 2m_{\pi^\pm})/3$ is the average pion mass. The function $T_\mu(r)$ matches the form from the OPE in Eq. (30) which means that $T_\mu^2(r)$ has the range of a two-pion force. The function $W(r)$ is of the Wood-Saxon form,

$$W(r) = \left[1 + e^{(r-r_0)/a} \right] . \quad (33)$$

Finally, the parameters I, P, Q, R are tuned to experiment, with some constraints.

This is the general structure that provides the intuition behind the choices made for the construction of the AV18 potential: a OPE terms is considered explicitly, the remaining intermediate- and short-range part is described by a combination of three operators $\mathbf{1}, \mathbf{L}, \mathbf{S}_{12}, \mathbf{L} \cdot \mathbf{S}, (\mathbf{L} \cdot \mathbf{S})^2$ whose r -dependent coefficients are different for various spin-isospin channels. We will now look at it in an alternate form which shows: a) why it's said to contain 18 operators and b) how it's connected to the rest of the Argonne potential family, AV4, AV8, AV14, etc.

First, we note that Eqs. (31) and (32) are essentially projections in spin-isospin channels of presumably more general ones. Formally, these projections are done using spin-isospin projection operators:

$$\Pi_\eta^j = \frac{1}{2} (1 + \eta P^j) \quad (34)$$

where $j = s, t$ for spin and isospin projection respectively, $\eta = -, +$ for singlet (odd) and triple (even) channels respectively, and P^j is a j -exchange operator that exchange the spin or isospin indices defined as

$$P^j = \frac{1}{2} \left(1 + \gamma^{(1)} \gamma^{(2)} \right) \quad (35)$$

for $j = s, t$ corresponding to $\gamma = \sigma, \tau$, the Pauli matrices acting on the spin or isospin spaces. Hence, the projection in a spin-isospin channel actually contains hidden spin and isospin operators. Factoring those in, and without a

projection, the potential between two particles i and j , where these indices are shorthand for $i = \{\mathbf{r}_i, s_{zi}, s_{zi}\}$ and similarly for j , can be written as

$$V_{ij} = \sum_{p=1}^{18} V_p(r_{ij}) O_{ij}^p . \quad (36)$$

These 18 operators give the name to the potential and they are numbered such as the first 14 correspond to the charge-independent ones that define the AV14:

$$\{O_{ij}^p\}_{p=1}^{14} = \{\mathbf{1}, \boldsymbol{\tau}_i \cdot \boldsymbol{\tau}_j, \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j, (\boldsymbol{\tau}_i \cdot \boldsymbol{\tau}_j)(\boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j), \mathbf{S}_{ij}, \mathbf{S}_{ij}(\boldsymbol{\tau}_i \cdot \boldsymbol{\tau}_j), \mathbf{L} \cdot \mathbf{S}, \mathbf{L} \cdot \mathbf{S}(\boldsymbol{\tau}_i \cdot \boldsymbol{\tau}_j), \quad (37)$$

$$\mathbf{L}^2, \mathbf{L}^2(\boldsymbol{\tau}_i \cdot \boldsymbol{\tau}_j), \mathbf{L}^2(\boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j), \mathbf{L}^2(\boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j)(\boldsymbol{\tau}_i \cdot \boldsymbol{\tau}_j), (\mathbf{L} \cdot \mathbf{S})^2, (\mathbf{L} \cdot \mathbf{S})^2(\boldsymbol{\tau}_i \cdot \boldsymbol{\tau}_j)\} . \quad (38)$$

The remaining 4 operators administer the CIB,

$$\{O_{ij}^p\}_{p=15}^{18} = \{\mathbf{T}_{ij}, \mathbf{T}_{ij}(\boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j), \mathbf{T}_{ij}\mathbf{S}_{ij}, (\tau_{zi} + \tau_{zj})\} , \quad (39)$$

where in analogy to the tensor operator $\mathbf{S}_{ij}$, the isotensor operator is defined as $\mathbf{T}_{ij} = 3\tau_{zi}\tau_{zj} - \boldsymbol{\tau}_i \cdot \boldsymbol{\tau}_j$.

4.2 Three-nucleon forces

The idea of many-nucleon forces dates back to almost a century ago when Wigner predicted their existence. The dire need for them to describe the ground-state of nucleonic matter from first principles was only recognized in the beginning of this century. The widely accepted evidence included the underbinding of the triton by all realistic 2NFs and the overbinding of nuclear matter. These issues were subsequently solved by appending highly accurate 2NFs by a two-pion exchange term employing various implementations and tuning strategies. In this section we will look at this very successful term: the Fujita-Miyazawa term.

4.2.1 The Fujita-Miyazawa term

In 1957 Fujita and Miyazawa prescribed the three nucleon force experienced by three nucleons where one nucleon emits a pion which is subsequently scattered (absorbed and re-emitted or fully scattered) and absorbed by a third nucleon. This process can cause the second nucleon to get excited to its lowest excited state, $\Delta(1232)$. Two things might be clear already: a) this is a genuine 3NF, that is there is no way to describe this interaction as the interaction of two nucleons at a time interacting with 2NFs and b) this force arises from neglecting the nucleons internal structure and treating it as a point particle: it is related to the nucleons internal structure, i.e., its excitations. The form of the Fujita-Miyazawa term in momentum space and up to multiplicative constants, later also doubted, two-pion exchange (TPE) 3NF, is

$$V_{\text{TPE}}(\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3) = \sum_{i \neq j \neq k} \frac{(\boldsymbol{\sigma}_i \cdot \mathbf{q}_i)(\boldsymbol{\sigma}_j \cdot \mathbf{q}_j)}{(\mathbf{q}_i^2 + m_\pi^2)(\mathbf{q}_j^2 + m_\pi^2)} \sum_{\alpha\beta} F_{ijk}^{\alpha\beta} \tau_i^\alpha \tau_j^\beta , \quad (40)$$

where V_{TPE} should be understood as a spin-isospin operator depending on the momentum exchange of particle $i = 1, 2, 3$, $\mathbf{q}_i = \mathbf{p}_i - \mathbf{p}'_i$, where $\mathbf{p}_i(\mathbf{p}'_i)$ is the initial (final) momentum of particle i , and

$$F_{ijk}^{\alpha\beta}(a, b, d) = \delta^{\alpha\beta} [a + b\mathbf{q}_i \cdot \mathbf{q}_j] + d \sum_{\gamma} \epsilon^{\alpha\beta\gamma} \tau_k^\gamma \boldsymbol{\sigma} \cdot (\mathbf{q}_i \times \mathbf{q}_j) , \quad (41)$$

where a , b , and d describe the s -wave and p -wave πN scattering. For example, in the original formulation it was assumed that the p -wave scattering happens in the $S, T = 3/2, 3/2$ channel which are the quantum numbers of the $\Delta(1232)$ excited state and that leads to $b = 4d$.

Since this force involves a pion exchanges it is of somewhat long range seen explicitly by the denominator of Eq. (40) that contains the propagator of the pion which when Fourier transformed to coordinate space tells of a range of each pion exchange of $e^{-\mu r}/(\mu r)$ with $\mu = m_\pi c/\hbar$. However being a 3NF, this force requires three nucleons coming within this range, or in other words, there are two pion propagators in the denominator of Eq. (40). So these forces will become important at higher nucleon densities.

It was right away shown by Fujita and Miyazawa that this type of force would produce an attraction for few nucleon systems and a repulsion in many nucleon systems. The few-nucleon estimation was done for the triton with a simple SD wavefunction (see first Lecture) and assuming that the three nucleons are arranged in an equilateral triangle with side ~ 1.3 fm, estimated by the Coulomb energy of He^3 . The many-nucleon estimation was similarly done for a Fermi gas model.

The Argonne family of potentials used this force appended by a repulsive spin-isospin remainder term V_R and tuned the relative strengths of V_{TPE} and V_R . Initially it was combined with AV18 and the two parameters were tuned to fix the binding energy of the triton and the ground-state energy of nuclear matter at saturation density. This produced a very good description of nuclei up to $A = 8$.

5 Chiral two-nucleon and three-nucleon forces

5.1 Introduction

The phenomenological models have loose connections to the underlying physics of the nucleon-nucleon interaction. Reiterating the points already discussed, the nucleon-nucleon interaction can be seen as a residual interaction of the quarks that compose them and the various models for it either try to describe it as one (or more) meson exchanges or more-or-less purely phenomenological fit of more-or-less all possible operators consistent with the expected symmetries of a nucleon-nucleon potential. We have already discussed the restrictions of either approach. Ideally one would like something that has a tighter connection to the underlying physics of quarks and gluons, that is, a connection to QCD. However, QCD is a non-perturbative theory and so direct treatments of quarks and their bound states on equal footing is currently not available. Chiral effective field theory (ChEFT) is an alternative approach following the philosophy of EFTs which constructs a theory that observes all relevant symmetries of the underlying theory. The original statement by Weinberg was: “If one writes down the most general possible Lagrangian, including all terms consistent with assumed symmetry principles, and then calculates matrix elements with this Lagrangian to any given order of perturbation theory, the result will simply be the most general possible S-matrix consistent with analyticity, perturbative unitarity, cluster decomposition, and the assumed symmetry principles.” Doing this to all available rigour involves several QFT techniques. We will go an alternative way: we will explain the basics, justify the assumptions, and survey the outcomes of the approach.

Let's first put the whole enterprise of developing an EFT for the nuclear interaction in simpler terms. The centrepiece in the development of any EFT is the *separation of scales*. This means if a system is described by two largely separated scales, say a small one Q and a large one M , then properties of the system can be written in terms of powers of the dimensionless quantity Q/M and the smaller the ratio the fewer terms are needed for a good description (i.e., the faster such a series will converge). Then the properties associated with energy less than the hard scale M will be successfully described while properties associated with energy larger than the hard scale are “baked-in” expansion coefficients that multiply the successive terms. These coefficients are named Low Energy Constants (LECs) and in principle they must be determined from a more complete theory. In our case this would be Lattice QCD that attempts the solution of QCD from first principles. However, in practice LECs are determined either by calculations from other EFTs that are valid for higher energies (EFT matching) or by experiment.

This approach has a few other attractive qualities. First it is systematically improvable: we can always include more terms and the description of a given quantity will get better. Second, two- and many-body forces are treated on equal footing and one doesn't need to be uncomfortable by matching terms that were developed separately. For all this enterprise to work it is crucial that the description developed satisfies the symmetry properties of the full theory

that describes the system. Now let's apply these concepts to few-nucleon systems. First, what are the soft and hard scales? Looking at the spectrum of quark bound states, there are immediate candidates: there is a large gap between the pion mass ($m_\pi \approx 140$ MeV) and the mass of the vector mesons $m_\rho \approx 770$ MeV and $m_\omega \approx 782$ MeV, also known as the chiral symmetry breaking scale. Hence we could identify $Q = m_\pi$ and $M = m_\rho$. This is practical but also physical. The small pion mass is a consequence of the nearly massless quarks in the QCD Lagrangian. Indeed, if the quarks were massless, the QCD Lagrangian would possess chiral symmetry which when broken spontaneously would give rise to its Goldstone boson, the pion; this too would be massless in this case. However, the quarks are not massless but their masses are small. This means that the chiral symmetry is explicitly broken in the QCD Lagrangian and the pion acquires a mass; this too is small. Therefore the existence of the soft scale can be traced back to the breaking of the chiral symmetry in the QCD Lagrangian. Further, identifying the soft scale with the pion mass explains the importance of the OPE term in all phenomenological potentials and provides some predictions: its a general property of Goldstone bosons that their interactions vanish for vanishing external momenta. This means that πN interactions are constrained to be not very large. This further means that many-nucleon interactions, which will by necessity be dominated by many-pion exchanges, are also expected to be small. This last point we'll revisit later as well.

Before diving into the specifics of the construction of chiral EFT interactions, note that sometimes the soft scale is also said to be the nucleon momentum exchange which in principle is distinct from the pion mass. It can be estimated that these two scales are of the same magnitude in low-energy processes as follows: the momentum exchange q can be estimated as the inverse of the average interparticle spacing which in a light nucleus can be estimated as $q \sim \sqrt{\langle r^2 \rangle_{\text{ch}}}$. For the triton this gives a scale similar to the pion mass.

We are now well equipped to understand the construction of chiral EFT interactions. The most general Lagrangian consistent with QCD's symmetries is written down and amplitudes from Feynman diagrams are calculated and organized according to the power $(Q/\Lambda_\chi)^\nu$ that they carry. As mentioned above, Q could be a generic nucleon momentum or a pion mass while $\Lambda_\chi \sim 1$ GeV is the chiral symmetry breaking scale. Assigning this power is called *power counting* and the most widely used prescription goes by the name of naive dimensional analysis or Weinberg power counting. According to the prescription a Feynman diagram contributes a force of order $(Q/\Lambda_\chi)^\nu$ with

$$\nu = -2 + 2A - 2C + 2L + \sum_i \Delta_i, \quad \text{with } \Delta_i = d_i + \frac{n_i}{2} - 2 \quad (42)$$

where A is the number of nucleons involved, C is the number of the diagram's separately connected pieces, L is the number of loops and the sum over i runs over all the diagram's vertices with d_i the number of derivatives or pion mass insertions and n_i the number of nucleon fields (or legs) at the vertex i . Note that $\nu \geq 0$ and so the chiral expansion converges (i.e., there are no terms proportional to the inverse of a nucleon's momentum). Thus at a given order $(Q/\Lambda_\chi)^\nu$ only a finite number of terms contribute and the effect of the neglected terms can be estimated as $\mathcal{O}((Q/\Lambda_\chi)^{\nu+1})$. We will now examine the two-body and three-body forces that arise up to next-to-next-to-leading-order (N²LO).

	NN	3N	4N
LO $\mathcal{O}(Q^0/\Lambda^0)$	 [151,152] 2	—	—
NLO $\mathcal{O}(Q^2/\Lambda^2)$	 [164,165] 7	 [192,194] [166-169]	—
N ² LO $\mathcal{O}(Q^3/\Lambda^3)$	 [164,165] 0	 [194] [167,170] 2	—
N ³ LO $\mathcal{O}(Q^4/\Lambda^4)$	 [2000-2002] [179-182] 12	 [2008-2011] [183-185] 0	 [2006] [186] 0
N ⁴ LO $\mathcal{O}(Q^5/\Lambda^5)$	 [2015] [188,189] 0	 [2011-] [190-192] 13+	 ?

5.2 Two-nucleon forces

For two nucleon forces naive dimensional power counting in Eq. (42) collapses to ($A = 2, C = 1$)

$$\nu = 2L + \sum_i \Delta_i . \quad (43)$$

Currently two-nucleon forces have been developed up to N⁵LO ($\nu = 6$) and they provide an excellent description of the two-nucleon system, that is, they describe the two-nucleon scattering phase shifts up to very high precision. Presenting all orders up to N⁵LO is redundant and we will focus on the first three which will allow us to understand the pattern and the obstacles involved.

5.2.1 LO, NLO forces, and N²LO forces

The lowest-order approximation to the nuclear force in ChEFT is the represented by the leading order (LO, $\nu = 0$). From Eq. (43) we see that this is obtained in diagrams with either no loops ($L = 0$), and one vertex with no derivatives or pion masses and two nucleon fields ($d_1 = 0, n_1 = 4$) or diagrams with no loops ($L = 0$) and two vertices with two nucleon fields and one pion mass each. These two types of contributions describe contact terms (two of them) and a OPE term, respectively. The existence of an OPE at leading order corroborates the idea that this type of term is one of the most important ones. While a bad approximation to the nuclear force, the LO force has some essential elements: the OPE that provides the tensor force (essential to describe the deuteron) and explains the NN scattering in peripheral waves of very high orbital angular momentum, the contacts provide a short- and intermediate-range interaction, which is however somewhat crude. Note that the leading order is the only order that does *not* have multi-pion exchanges.

The subleading contribution to the two-nucleon force comes at $\nu = 2$ since the $\nu = 1$ order cancels entirely from parity conservation and time-reversal invariance. Since $\mathcal{O}((Q/\Lambda_\chi)^{\nu+1})$ can be used to estimate the order-of-magnitude contribution of the neglected terms, the absence of $\nu = 1$ speaks to an increased accuracy of the LO compared to what would be naively thought (i.e., the missing contributions are $\mathcal{O}((Q/\Lambda_\chi)^2)$). At NLO the leading contribution of the two-pion exchange terms enter as the first multi-pion terms. As mentioned before, this term is essential for the description of the intermediate range force, but it's weakened at this order. Seven more contacts appear with their corresponding LECs and more one-pion exchanges appear as well renormalizing (i.e., shifting) the strength of the one-pion exchange from LO. As we will discuss in the next section, in principle, the first three-nucleon forces appear at this order but they cancel out.

Finally, at N²LO more two-pion exchange terms enter. These contain contributions from the $\Delta(1232)$ isobar and provide the necessary strength to the two-pion exchange to describe intermediate range forces. As you might guess, the appearance of contributions from $\Delta(1232)$ hint that three-nucleon-forces might be of this order too: the Fujita-Miyazawa term describes contributions from the $\Delta(1232)$ isobar. This intuition is correct as we will discuss below. No new contacts enter at this order, however, the first relativistic corrections appear but these cancel at the center-of-mass frame where most many-body calculations are done. Coincidentally, N²LO is the highest order at which the chiral forces can be formulated as purely local.

Bringing all orders together, the two-nucleon potential from chiral EFT up to N²LO is

$$V_{\text{LO}} = V_{\text{ct}}^{(0)} + V_{1\pi}^{(0)} \quad (44)$$

$$V_{\text{NLO}} = V_{\text{LO}} + V_{\text{ct}}^{(2)} + V_{1\pi}^{(2)} + V_{2\pi}^{(2)} \quad (45)$$

$$V_{\text{N}^2\text{LO}} = V_{\text{NLO}} + V_{1\pi}^{(3)} + V_{2\pi}^{(3)} \quad (46)$$

5.3 Three-nucleon forces

A major attractive feature of chiral EFT is that it treats two- and many-body forces on equal footing. This puts many conceptual matters to rest but it also provides a justification for the apparent suppression of many-body forces: they enter at higher order in the chiral expansion. Indeed, looking at Eq. (42), an m -nucleon force in a $A \geq m$ nucleon system gives

$$\nu_m^* = 2m - 4 + 2L + \sum_i \Delta_i \quad (47)$$

We can determine the lowest possible ν_m^* by noting that $L > 0$ (a negative number of loops is nonsensical) and that for interactions allowed by chiral symmetry $\sum_i \Delta_i \geq 0$. So the lowest order that a m -body force appears at is

$$\nu_m = 2m - 4 \quad (48)$$

That is two-nucleon forces start at $\nu_2 = 0$, that is, at LO. The first three-nucleon force would enter at $\nu_3 = 2$, but they happen to cancel there, so they enter at $\nu_3 + 1 = 3$, that is, at N²LO. Four-nucleon forces enter at $\nu_4 = 4$, that is, at N³LO, and so on. This provides a hierarchy to the many-body forces that was otherwise assumed. Moreover, it tells of a suppression of order $(Q/\Lambda_\chi)^2$ from one leading many-body force to the next, except for the three-nucleon force, whose would-have-been leading contribution cancels at $\nu = 2$. Here we will look at the actual leading order contributions of the three-nucleon force.

5.3.1 Leading: N²LO

As was mentioned in 5.2.1, the first contribution to the two-pion exchange by the $\Delta(1232)$ isobar enters at N²LO which hints towards the existence of three-nucleon forces at that order which can also depend on the same mechanism. Indeed at this order the first three-nucleon forces enter in the form of the Fujita-Miyazawa two-pion exchange term, a one pion exchange term, and a contact. These three terms have the form

$$V_{\text{ct}}(\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3) = \frac{c_E}{2f\pi^4\Lambda_\chi} \sum_{i \neq j \neq k} \boldsymbol{\tau}_i \cdot \boldsymbol{\tau}_j \quad (49)$$

$$V_{\text{OPE}}(\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3) = -\frac{c_{DGA}}{8f_\pi^4\Lambda_\chi} \sum_{i \neq j \neq k} \frac{\boldsymbol{\sigma}_j \cdot \mathbf{q}_j}{\mathbf{q}_j^2 + m_\pi^2} (\boldsymbol{\tau}_i \cdot \boldsymbol{\tau}_j) (\boldsymbol{\sigma}_i \cdot \mathbf{q}_j) \quad (50)$$

$$V_{\text{TPE}}(\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3) = \left(\frac{g_A}{2f_\pi}\right)^2 \frac{1}{2} \sum_{i \neq j \neq k} \frac{(\boldsymbol{\sigma}_i \cdot \mathbf{q}_i)(\boldsymbol{\sigma}_j \cdot \mathbf{q}_j)}{(\mathbf{q}_i^2 + m_\pi^2)(\mathbf{q}_j^2 + m_\pi^2)} \sum_{\alpha\beta} F_{ijk}^{\alpha\beta} \tau_i^\alpha \tau_j^\beta, \quad (51)$$

with

$$F_{ijk}^{\alpha\beta}(c_1, c_3, c_4) = \delta^{\alpha\beta} \left[-\frac{4c_1 m_\pi^2}{f_p i^2} + \frac{2c_3}{f_\pi^2} \mathbf{q}_i \cdot \mathbf{q}_j \right] + \sum_\gamma \frac{c_4}{f_\pi^2} \epsilon^{\alpha\beta\gamma} \tau_k^\gamma \boldsymbol{\sigma} \cdot (\mathbf{q}_i \cdot \mathbf{q}_j) \quad (52)$$

The LECs in the TPE exchange also enter in the two-nucleon force but they can also be connected to πN scattering. As such, they don't carry pure three-nucleon information and they are usually predetermined at this order. Similarly the c_D LEC also enters in the two body currents, that is, the response of the two-body system to external probes, and as such it can also be determined from β -decay. However, the LECs c_E have to be determined by genuine three-body physics.

6 Faking nucleon forces with cold atoms: Feshbach resonances and the unitarity regime

One thing should be clear from the preceding discussion: the nuclear force is very complicated. Nuclear experiments are routinely overcoming records in precision, those are available exclusively for finite nuclei whose properties are often very different from those of extended nucleonic matter. The latter is a simpler system, free of finite-size effects yet realistic as it naturally arises within compact stars. The properties of nucleonic matter cannot be studied directly in the lab. However, the presence of the strong nuclear force among its constituents which immensely complicates its theoretical description offers a way out on the question of experimental probes. This goes along the following logic. There is only so much two particles can attract before creating a free-space bound state. Since the two-particle energy switches sign once the bound-state forms, on the verge of the bound state forming, the two strongly attracting particles should form a state of vanishing binding energy. Similarly, the (inverse) scattering length of the two-body scattering switches sign so on the verge of a bound state forming the (inverse) scattering length should (vanish) diverge. If the interaction that causes this situation is zero-range then this scenario has no length scales coming from the two-body interaction:

$$1/a = 0 \quad \text{and} \quad r_e = 0 . \quad (53)$$

This regime is called the *unitarity regime* and the system the unitary gas. Since all length scales of the interaction have dropped out one can expect that the details of the potential do not matter: the same universal behaviour is expected from all unitary systems. Of course, most systems in nature are not fine-tuned at unitarity. At the same time, of course, the conditions in Eq. (53) are not complete as they don't represent relations among unitless quantities. These two issues are related and one can more appropriately define the unitarity regime as

$$k_F/a = 0 \quad \text{and} \quad k_F r_e = 0 . \quad (54)$$

In other words, any fermionic system with strong interactions (a not tiny) and short-range interactions (r_e not large), approaches a unitary gas as sufficiently low density. As such, the unitarity regime can be seen as the low-density limit of all Fermi gases interacting with a strong short-ranged interaction. Nucleons, and especially neutrons, are major examples in this category, and dilute neutron matter should exhibit behavior close to a unitary Fermi gas. However, this is a two-way street: unitary gases can then be used to probe effects in neutron matter.

The connection described above exists at low energies and was first introduced by Bertsch: the physics of the unitary Fermi gas is universal with a single length scale coming from the density. To the extent that the effective range of the neutron-neutron interaction can be neglected, its large scattering length places neutron matter close to a unitary Fermi gas at low densities. This makes emulating neutron matter in the lab possible using fermionic gases with tunable interactions. A prime example of such systems are cold-atomic gases where the interaction can be tuned using Feshbach resonances.

A Feshbach resonance is a way to (“artificially”) increase the scattering length of the interaction between cold atoms. For bosonic particles there are losses because Feshbach resonances couple the atomic Hilbert space to a resonant molecular state which is vibrationally highly excited, but these losses are heavily suppressed for fermions. Wide use started in 2002 (late 2001), while in 2003 the first Feshbach molecules were created, i.e., a BEC condensation of molecules comprised of fermions.

Feshbach resonances occur as a 2-body bound state in the interatomic potential is tuned into resonance with the energy of two colliding atoms. This is done via an applied magnetic field if the magnetic moment of the bound state differs from that of the two unbound atoms. There are certain conditions where the population of the bound state (“closed channel”) can be eliminated, and the physics is reduced to potential scattering (“broad Feshbach resonance”).

The characteristic size of a fermion dimer formed by a two-body interaction of scattering length a is $r_{\text{dim}} \sim a$. These can relax into more deeply bound states with a size $r' \sim r_0$, where r_0 is the van der Waals range $r_0 = 50 - 100a_0$ and $a_0 \approx 5.3 \times 10^4$ fm the Bohr radius. This is considered a loss in the signal: we want the dimers of large size (a) not the tightly bound molecules of small size (r_0). However, this is a vibrational relaxation: the energy lost is transferred to a third fermion nearby. For that to happen three fermions need to be within a distance of r_0 from each other. Since two of them (at least) will have opposite spins, their wavefunction should be antisymmetrized. The antisymmetrized wave function has a node at zero separation ($\psi(r=0) = 0$), and it grows as r/a at small r . So, the probability of finding three fermions at a distance r_0 (and the rate of relaxation) is suppressed by a factor of $(r_0/a)^p$ for some p and $r_0/a \ll 1$ (large scale difference). This is a manifestation of the Pauli principle. Hence we can think of it as an extra repulsive interaction making the more tightly bound state less energetically favorable (i.e., Pauli repulsion).

The value of Feshbach resonances for nuclear physics lays in their ability to “emulate” strongly correlated matter interactions, at low energies, in the lab. This is because the interparticle potential’s low energy expansion is defined exclusively by the scattering length a and effective range r_e . Hence, one can use Feshbach resonances to create effective interparticle potentials that resemble the nuclear one to sub-leading order in the effective range expansion.